

A Four-State Laboratory for Causal Dissonance, Insight, and Strategic Learning

Research note

July 2026

Abstract

This note develops a deterministic two-firm, four-state model of causal dissonance and experience-triggered awareness change. Firms initially cannot formulate a customer-contingent causal theory. Heterogeneous private beliefs induce different committed offering rules, and the resulting experience empties the active subset of their current model space. A successful inquiry episode is represented by two distinct products: a minimally restorative awareness refinement, followed by an associated causal formulation. The 2×2 customer-state geometry makes the central difficulty explicit: the same dissonant experience supports four minimal refinements and two equally simple factor-based explanations. No truth-blind inquiry rule can infer the correct factor from those data alone. Robustly accommodating both observationally equivalent causal worlds requires complete awareness, while a non-exhaustive technology must accept fallibility. Because a restorative refinement fits its generating experience by construction, retrospective fit carries no discriminating information: representation-relative certainty exceeds selection-adjusted warrant by a computable gap that pooled experience never narrows and a single fresh differentiated observation closes. Expressing the comparison behind that gap requires complete awareness, so no minimally refined firm can audit its own formulation from within. The model also shows that strategic action controls the diagnostic content of experience. Shared false formulations induce pooled actions and can become self-confirming; heterogeneous formulations generate revealing competition, learning, and asymmetric value capture. These pathologies are knife-edge properties of perfectly forgiving demand: with any outside option, pooled misservice destroys observable value, and consistency together with subjective optimality forces every recurring state to be correctly served—payoff-relevant self-confirming traps, permanent warrant gaps, and zero-leverage locks all require demand to absorb misservice completely. An inquiry-participation layer then prices the question that dissonance raises: a delabeled leverage statistic, computed from retained episodes without naming any candidate representation, determines when costly inquiry is subjectively justified. Pooled histories carry zero leverage, so the awareness trap survives even cheap and infallible inquiry, while differentiated defeat concentrates both the evidence and the incentive to inquire in the losing firm. The note separates representation generation, causal formulation, Bayesian assessment, reflective judgment, and strategic decision and interprets that separation in relation to Lonergan’s account of inquiry and insight.

1 Purpose and status of the example

The note is a formal laboratory rather than a complete paper model. Its purpose is to determine what can be established in the smallest environment that permits several qualitatively different awareness refinements: rows, columns, diagonals, singleton exceptions, and finer combinations. The model is designed to answer five questions.

- (i) How can strategic experience reveal that every currently expressible causal theory is inadequate?
- (ii) What restrictions does the experience place on a satisfactory new representation?
- (iii) Why do those restrictions direct inquiry without logically determining its answer?
- (iv) Where does fixed-space Bayesian learning stop, and where do representation production and subsequent judgment begin?
- (v) How do the firms' resulting strategies affect whether a false formulation will subsequently be exposed?
- (vi) When is costly inquiry subjectively justified, and what represented statistic can price a question whose possible answers the firm cannot yet formulate?
- (vii) How much confidence does the generating experience actually license in a produced formulation, and which agents can compute that warrant?
- (viii) Which epistemic pathologies survive when misservice destroys value rather than merely reallocating it?

Four modeling distinctions are maintained throughout. An *awareness partition* determines which customer distinctions a firm can represent. A *causal theory* assigns a customer response to every cell of that partition. A *causal formulation* combines a newly available partition with an expressible theory. A *strategy* maps represented customer states to product architectures. None of these objects is identified with insight itself; the mathematics represents the output of a successful insight-producing process.

2 Notation

Symbol	Meaning
$N = \{1, 2\}$	Set of firms.
$A = \{0, 1\}$	Set of architectures and firm actions.
$X = \{x_{00}, x_{01}, x_{10}, x_{11}\}$	Analyst's finite customer-state space.
$d : X \rightarrow A$	Objectively preferred architecture.
u_i	Firm i 's objective payoff function.
β	Demand retention under pooled misservice; $\beta = 1$ is the baseline.
$\Pi(X)$	Set of all partitions of X .
$\mathcal{P}_i \in \Pi(X)$	Firm i 's awareness partition.
$M(\mathcal{P}_i)$	Set of causal theories expressible under $\mathcal{P}_i$.
$m \in M(\mathcal{P}_i)$	An individual causal theory.
$\mu_i \in \Delta(M(\mathcal{P}_i))$	Firm i 's belief over its expressible theories.
$\sigma_i : \mathcal{P}_i \rightarrow A$	Firm i 's product policy.
H	Set of analyst-level histories.
h	Analyst's record of customer states, actions, and payoffs.
$e_i(h; \mathcal{P}_i)$	Firm i 's deliberative record of retained episode tokens.
$D(h) \subseteq X \times A$	Causal labels revealed by differentiated actions.
$\hat{M}(\mathcal{P}, D)$	Theories under $\mathcal{P}$ consistent with D .
$E(D)$	Conflict-edge set induced by D .
$\mathcal{R}(\mathcal{P}, D)$	Set of minimally restorative refinements.
$\kappa(\mathcal{P})$	Coordinate complexity of a binary partition.
$\phi_i : \Pi(X) \times H \rightarrow \Pi(X)$	Analyst-level representation-producing technology.
$\mathcal{I}(\mathcal{P}, D)$	Complexity-filtered restorative correspondence.
$\bar{\mu}_i$	Representation-relative coherence distribution.
ε_i	Firm i 's binary inquiry election.
γ_i	Firm i 's inquiry activation cost.
θ_i	Firm i 's delabeled inquiry-success weight.
$L_i(h)$	Disagreement set: retained labeled episodes contradicting the committed policy.
$B_i(h)$	Retained-record leverage of inquiry.
$\mathcal{D}$	Candidate objective laws in a reflective assessment.
ν	Reflective prior over $\mathcal{D}$.
$W(m' h)$	Selection-adjusted warrant of formulation m' .
$G(m' h)$	Warrant gap: coherence probability minus warrant.
$\mathcal{P}^r, \mathcal{P}^c, \mathcal{P}^\times$	Row, column, and diagonal partitions.
$\mathcal{P}^{jk}$	Partition isolating x_{jk} from the other states.
$\mathcal{P}^{\text{full}}$	Partition of X into four singleton cells.

Capital Roman letters denote sets, lowercase Roman and Greek letters denote elements, parameters, and functions, calligraphic letters denote sets of sets, and blackboard-bold notation is

reserved for canonical sets. For any finite set Z , $\Delta(Z)$ denotes the set of probability distributions on Z .

3 Objective environment and timing

3.1 Customer states and objective fit

There are two firms,

$$N = \{1, 2\},$$

and two available architectures,

$$A = \{0, 1\}.$$

The analyst's customer-state space is

$$X = \{x_{00}, x_{01}, x_{10}, x_{11}\}.$$

It is convenient to display the states as a matrix:

	$c = 0$	$c = 1$
$r = 0$	x_{00}	x_{01}
$r = 1$	x_{10}	x_{11}

The two coordinates are analyst-level descriptions of potentially usable customer distinctions. Their status in the firms' awareness is specified in [section 4](#).

The objectively preferred architecture is

$$d : X \longrightarrow A, \quad d(x_{rc}) = r. \tag{1}$$

Thus the first coordinate is causally relevant and the second is not:

	$c = 0$	$c = 1$
$r = 0$	0	0
$r = 1$	1	1

If the firms offer different architectures, the customer buys from the firm offering $d(x)$. If they offer the same architecture, they split the unit market, even if another architecture would have served the customer better. Firm i 's payoff is

$$u_i(a_i, a_j, x) = \begin{cases} \frac{1}{2}, & a_i = a_j, \\ 1, & a_i \neq a_j \text{ and } a_i = d(x), \\ 0, & a_i \neq a_j \text{ and } a_j = d(x). \end{cases} \tag{2}$$

The payoffs sum to one. A customer state can equivalently be interpreted as a unit market segment, which avoids interpreting the one-half outcome as random tie-breaking for an indivisible customer.

3.2 Batch commitment

The opening experience occurs within a product generation. At the start of a generation, each firm simultaneously commits to a contingent offering or configuration rule. A finite batch of customer states is then served under those rules. Firms observe the customer episodes, actions, and payoffs, but they cannot revise the committed rule until the batch ends. Initial coarse awareness makes the rule constant; refined awareness permits customer- contingent implementation. Updating and inquiry take place only between generations.

Assumption 3.1 (Diagnostic batch). *The initial batch consists of the two states x_{00} and x_{11} . Both are served under the policies chosen before either outcome is observed.*

The batch commitment is substantive. With deterministic period-by-period updating and immediate redesign, the first differentiated observation would collapse both firms' active sets onto the same constant theory. They would then choose the same architecture at the second state, suppressing the causal signal needed for the focal dissonance.

4 Awareness and expressible causal theories

Definition 4.1 (Awareness partition). *An awareness partition $\mathcal{P}_i \in \Pi(X)$ determines the customer distinctions firm i can use in causal theories and prospective product policies. A policy is a function*

$$\sigma_i : \mathcal{P}_i \longrightarrow A.$$

The action taken at x is $\sigma_i(C)$, where $C \in \mathcal{P}_i$ is the cell containing x . For brevity, $\sigma_i(x)$ denotes this induced statewise action.

For any $\mathcal{P} \in \Pi(X)$, let $C_{\mathcal{P}}(x)$ denote the unique cell of $\mathcal{P}$ containing x .
Initially, both firms have the coarse awareness partition

$$\mathcal{P}_i^0 = \{X\}. \quad (3)$$

A firm with this partition cannot formulate a causal theory, or commit to a policy, that distinguishes any pair of customer states.

Let n denote the claim that architecture is causally neutral. For any partition $\mathcal{P}$, define

$$M(\mathcal{P}) = \{m : X \longrightarrow \{0, 1, n\} : m(x) = m(x') \text{ whenever } x, x' \text{ lie in the same cell of } \mathcal{P}\}. \quad (4)$$

For $C \in \mathcal{P}$ and $m \in M(\mathcal{P})$, write $m(C)$ for the common value $m(x)$ on C , which is well defined by cell constancy. The interpretations are

$$m(x) = a : \text{architecture } a \text{ is preferred at } x, \quad m(x) = n : \text{architecture is neutral at } x.$$

The payoff predicted by m is

$$u_i^m(a_i, a_j, x) = \begin{cases} \frac{1}{2}, & a_i = a_j \text{ or } m(x) = n, \\ 1, & a_i \neq a_j \text{ and } m(x) = a_i, \\ 0, & a_i \neq a_j \text{ and } m(x) = a_j. \end{cases} \quad (5)$$

Under $\mathcal{P}_i^0$, the only expressible theories are the three constant functions

$$M(\mathcal{P}_i^0) = \{m^0, m^1, m^n\},$$

where $m^a(x) = a$ for every $x \in X$ and $a \in \{0, 1, n\}$. The true law d is not in $M(\mathcal{P}_i^0)$.

The partition is an analyst-level representation of the firm's subjective state space. At the level of deliberation and expressible belief, the firm has the quotient state space $X/\mathcal{P}_i$, whose elements are the cells of $\mathcal{P}_i$; it does not possess the master space X together with an unobserved choice among finer partitions. Every pre-insight belief, likelihood, deliberately entertained question, and policy is $\mathcal{P}_i$ -measurable. In particular, before awareness expands, the firm cannot name a finer partition, assign it a probability, or condition a policy on it.

Assumption 4.2 (Retention and recodability). *Customer episodes can be retained as particulars and retrospectively recoded after the firm acquires a new predicate. Once acquired, the predicate can also classify a future customer before the firm acts. Before acquisition, neither belief nor action can condition on that predicate.*

This assumption supplies the minimal distinction between experiencing a particular and possessing a projectible causal concept about it. Without some extra-deliberative access to retained particulars, a strict refinement could not arise endogenously from a subjective state space that contains only one state. Retaining a particular is not possession of an event that separates it from other customer states and cannot support a contingent policy before recoding. The representation-producing technology introduced below is a primitive of concept production: it can operate on retained particulars through latent comparison capacities, but those operations are not available to the firm's pre-insight deliberation.

5 Private beliefs and the opening product policies

Firm i has a private prior

$$\mu_i^0 \in \Delta(M(\mathcal{P}_i^0)).$$

The priors need not be common knowledge. For a general awareness cell $C \in \mathcal{P}_i$, write

$$p_i^a(C) = \mu_i(\{m \in M(\mathcal{P}_i) : m(C) = a\}), \quad a \in \{0, 1, n\}.$$

Let $v_i(a \mid C)$ denote firm i 's subjective expected payoff from action a at cell C , given its current causal belief and any conjecture about the rival's action.

Assumption 5.1 (Subjective optimality and commitment). *Whenever firm i has a well-defined current causal belief, it selects at the start of each generation an action maximizing $v_i(a \mid C)$ at every represented cell $C \in \mathcal{P}_i$ and commits to the resulting policy for the generation's batch. The carry-forward clause in [assumption 9.5](#) governs the exceptional state in which dissonance has made belief undefined and representation production has failed.*

Lemma 5.2 (Subjective dominance). *For either pure action of the rival, and therefore for any mixture over the rival's actions,*

$$v_i(0 \mid C) - v_i(1 \mid C) = \frac{1}{2}[p_i^0(C) - p_i^1(C)]. \quad (6)$$

Consequently, action 0 is subjectively strictly dominant on C if $p_i^0(C) > p_i^1(C)$, and action 1 is subjectively strictly dominant if the inequality is reversed.

Proof. Suppose first that the rival selects 0. Selecting 0 yields one-half under every theory. Selecting 1 yields zero under a theory labeling the cell 0, one under a theory labeling it 1, and one-half under a neutral theory. The expected-payoff difference is therefore

$$\frac{1}{2} - \left[p_i^1(C) + \frac{1}{2}p_i^n(C) \right] = \frac{1}{2}[p_i^0(C) - p_i^1(C)].$$

If the rival selects 1, action 0 yields $p_i^0(C) + \frac{1}{2}p_i^n(C)$ and action 1 yields one-half, giving the same difference. Linearity establishes the result against any mixture. $\square$

Assumption 5.3 (Heterogeneous initial causal beliefs).

$$\mu_1^0(m^0) > \mu_1^0(m^1), \quad \mu_2^0(m^1) > \mu_2^0(m^0).$$

Proposition 5.4 (Belief-induced differentiation). *Under assumptions 3.1, 5.1 and 5.3, firm 1 commits to architecture 0 throughout the opening batch and firm 2 commits to architecture 1.*

Proof. Under $\mathcal{P}_i^0$, each product policy must be constant on X . Lemma 5.2 and assumption 5.3 make action 0 strictly dominant for firm 1 and action 1 strictly dominant for firm 2. Batch commitment then holds the differentiated action profile fixed while both diagnostic states are served. $\square$

6 Revealed causal evidence and Bayesian closure

Let H be the set of finite analyst-level histories, with a typical history written

$$h = (x_s, a_{1s}, a_{2s}, u_{1s}, u_{2s})_{s=1}^t.$$

At the deliberative level, firm i represents this history under $\mathcal{P}_i$ as

$$e_i(h; \mathcal{P}_i) = (s, C_{\mathcal{P}_i}(x_s), a_{1s}, a_{2s}, u_{1s}, u_{2s})_{s=1}^t. \quad (7)$$

The index s retains an episode as a particular token; it is not a customer predicate on which a pre-insight theory or policy may condition. The analyst-level history also records the underlying customer state so that a newly produced representation can retrospectively recode the token.

When the actions differ, the payoff vector identifies the winning architecture. Let

$$D(h) = \{(x_s, a_{is}) : s \in \{1, \dots, t\}, i \in N, a_{1s} \neq a_{2s} \text{ and } u_{is} = 1\} \subseteq X \times A \quad (8)$$

be the analyst's recoding of causal labels attached to the retained episodes. Before refinement, the firm does not observe the ordered pair (x_s, a_{is}) as a proposition on the master space. It experiences two episode tokens in its same represented cell with the displayed actions and opposing payoffs. After refinement, assumption 4.2 permits the tokens to be recoded into the new cells. A pooled observation is omitted because its payoff vector is $(1/2, 1/2)$ under every objective and subjective causal law.

A revealed dataset is *functional* if

$$(x, a), (x, a') \in D \implies a = a'. \quad (9)$$

Every dataset generated by the deterministic objective law is functional.

For a proposed awareness partition $\mathcal{P}$, define the active model set

$$\widehat{M}(\mathcal{P}, D) = \{m \in M(\mathcal{P}) : m(x) = a \text{ for every } (x, a) \in D\}. \quad (10)$$

Definition 6.1 (Causal dissonance). *An awareness partition $\mathcal{P}$ exhibits causal dissonance at dataset D when*

$$\widehat{M}(\mathcal{P}, D) = \emptyset.$$

The expressible model set $M(\mathcal{P})$ does not disappear. Rather, no model in it accounts for all revealed causal evidence.

In the opening batch, equation (1) and proposition 5.4 imply

$$\overline{D} = \{(x_{00}, 0), (x_{11}, 1)\}. \quad (11)$$

No constant theory fits both observations, so

$$\widehat{M}(\mathcal{P}_i^0, \overline{D}) = \emptyset. \quad (12)$$

The immediate question has a determinate defect but not a determinate answer: what difference between x_{00} and x_{11} explains why different architectures fit them?

Lemma 6.2 (Endogenous revelation). *At a customer state x , the objective label $d(x)$ is revealed by the payoff vector if the firms choose different architectures. If they choose the same architecture, the payoff vector is independent of $d(x)$ and supplies no restriction on the active model set.*

Proof. With two architectures, differentiated actions imply that exactly one firm offers $d(x)$. The firm receiving payoff one identifies that architecture. Under pooled actions, equation (2) gives both firms one-half for either value of $d(x)$. $\square$

Lemma 6.3 (Fixed-space Bayesian closure). *Fix $\mathcal{P}$ and a prior $\mu \in \Delta(M(\mathcal{P}))$. Whenever Bayesian conditioning has a positive denominator, the posterior remains in $\Delta(M(\mathcal{P}))$. Every theory in posterior support and every optimal policy therefore remains constant within each cell of $\mathcal{P}$. If $\widehat{M}(\mathcal{P}, D) = \emptyset$ in the deterministic model, the Bayesian posterior is undefined and conditioning supplies no strict refinement of $\mathcal{P}$.*

Proof. Conditional on the realized customer-state and action design, define the deterministic likelihood

$$\ell(D \mid m, \mathcal{P}) = \mathbf{1}\{m \in \widehat{M}(\mathcal{P}, D)\}.$$

Whenever the denominator is positive, Bayes' rule gives

$$\mu(m \mid D, \mathcal{P}) = \frac{\mu(m)\ell(D \mid m, \mathcal{P})}{\sum_{m' \in M(\mathcal{P})} \mu(m')\ell(D \mid m', \mathcal{P})}.$$

Conditioning changes probability weights but not the domain on which those weights are defined. By equation (4), every element of $M(\mathcal{P})$ is constant within $\mathcal{P}$ -cells, so any posterior mixture and any policy optimized against it respects those cells. When every model assigns likelihood zero to the data, Bayes' denominator is zero. No operation in the conditioning formula introduces a partition outside the prior domain. $\square$

If the prior lacks full support, a nonempty active set can still receive total prior mass zero. The direction used here is the sharper one needed for causal dissonance: an empty active set necessarily gives a zero denominator under every prior.

Remark 6.4 (Algebraic interpretation). *Let the event algebra generated by $\mathcal{P}$ consist of all unions of its cells. Boolean operations and conditioning within that algebra cannot produce an event separating two states in the same cell. A strict awareness refinement therefore requires a representation-producing primitive that is not reducible to fixed-space Bayesian conditioning.*

7 Restorative inquiry and the conflict structure

For $\mathcal{P}, \mathcal{P}' \in \Pi(X)$, write

$$\mathcal{P} \prec \mathcal{P}'$$

when $\mathcal{P}'$ strictly refines $\mathcal{P}$: every cell of $\mathcal{P}'$ lies within a cell of $\mathcal{P}$, and the partitions are not equal.

Definition 7.1 (Restorative refinement). *Partition $\mathcal{P}'$ is restorative at $(\mathcal{P}, D)$ if*

$$\mathcal{P} \prec \mathcal{P}' \quad \text{and} \quad \hat{M}(\mathcal{P}', D) \neq \emptyset.$$

It is minimally restorative if no restorative $\mathcal{P}''$ satisfies

$$\mathcal{P} \prec \mathcal{P}'' \prec \mathcal{P}'.$$

Let $\mathcal{R}(\mathcal{P}, D)$ denote the set of minimally restorative refinements.

Minimality is defined in the refinement partial order. It does not imply that there is a unique coarsest answer.

Define the conflict-edge set

$$E(D) = \{ \{x, x'\} \subseteq X : (x, 0), (x', 1) \in D \text{ or } (x, 1), (x', 0) \in D \}. \quad (13)$$

A subset of X is conflict-free if it contains neither endpoint pair of any edge in $E(D)$.

Proposition 7.2 (Conflict characterization). *For any $\mathcal{P} \in \Pi(X)$ and functional revealed dataset D ,*

$$\hat{M}(\mathcal{P}, D) \neq \emptyset$$

if and only if every cell of $\mathcal{P}$ is conflict-free. Consequently, $\mathcal{P}'$ is restorative at $(\mathcal{P}, D)$ if and only if it strictly refines $\mathcal{P}$ and every cell of $\mathcal{P}'$ is conflict-free.

Proof. If a cell contains the endpoints of a conflict edge, the observations require a cell-constant theory to assign both zero and one to that cell, which is impossible. Conversely, suppose every cell is conflict-free. If a cell contains a revealed state, all revealed states in it have the same label; assign that label to the cell. Assign any value in $\{0, 1, n\}$ to an unlabeled cell. The resulting cell-constant function belongs to $\hat{M}(\mathcal{P}, D)$. $\square$

The conflict characterization generalizes beyond the focal example. The restorative correspondence identifies minimally refined partitions whose cells form proper colorings of the experienced conflict relation. It determines what a successful representation-producing process must accomplish without selecting a particular answer.

8 Complete four-state enumeration

The fifteen partitions of a four-element set have the following cell-size patterns.

Cell-size pattern	Number	Role in the example
4	1	Initial awareness
3 + 1	4	Singleton-exception refinements
2 + 2	3	Rows, columns, and diagonals
2 + 1 + 1	6	Three-cell refinements
1 + 1 + 1 + 1	1	Complete awareness
Total	15	

The three balanced two-cell partitions are

$$\mathcal{P}^r = \{\{x_{00}, x_{01}\}, \{x_{10}, x_{11}\}\}, \quad (14)$$

$$\mathcal{P}^c = \{\{x_{00}, x_{10}\}, \{x_{01}, x_{11}\}\}, \quad (15)$$

$$\mathcal{P}^\times = \{\{x_{00}, x_{11}\}, \{x_{01}, x_{10}\}\}. \quad (16)$$

For $j, k \in \{0, 1\}$, define the four singleton-exception partitions

$$\mathcal{P}^{jk} = \{\{x_{jk}\}, X \setminus \{x_{jk}\}\}. \quad (17)$$

These seven two-cell partitions are exactly the immediate refinements of $\mathcal{P}^0 = \{X\}$.

Proposition 8.1 (Four minimally restorative refinements). *For the discovery dataset $\bar{D}$ in equation (11),*

$$\mathcal{R}(\mathcal{P}^0, \bar{D}) = \{\mathcal{P}^r, \mathcal{P}^c, \mathcal{P}^{00}, \mathcal{P}^{11}\}. \quad (18)$$

The uniquely active theories under these partitions, written in the state order $(x_{00}, x_{01}, x_{10}, x_{11})$, are

$$m^r = (0, 0, 1, 1),$$

$$m^c = (0, 1, 0, 1),$$

$$m^{00} = (0, 1, 1, 1),$$

$$m^{11} = (0, 0, 0, 1).$$

Under the objective law in equation (1), $m^r = d$ and the other three formulations are false outside the discovery data.

Proof. By [proposition 7.2](#), a restorative partition must separate x_{00} from x_{11} . Exactly four two-cell partitions do so: rows, columns, the partition isolating x_{00} , and the partition isolating x_{11} . The diagonal partition and the singleton partitions isolating x_{01} or x_{10} leave x_{00} and x_{11} in the same cell. Every restorative partition with three or four cells strictly refines at least one of the four restorative two-cell partitions and is therefore not minimal.

In each of the four listed partitions, one cell contains the observation $x_{00} \mapsto 0$ and the other contains $x_{11} \mapsto 1$. Cell constancy therefore determines the displayed theory uniquely. Direct comparison with $d = (0, 0, 1, 1)$ establishes the final claim. $\square$

Proposition 8.2 (Multiplicity with one conflict). *Let C be the only non-conflict-free cell of the current partition, let $|C| = q \geq 2$, and suppose exactly one conflict edge has both endpoints in C . There are 2^{q-2} minimally restorative refinements that replace C by two cells and leave every other awareness cell unchanged.*

Proof. Fix the endpoint labeled zero in the first new cell and the endpoint labeled one in the second. Each of the remaining $q - 2$ states can independently be placed in either cell, producing 2^{q-2} binary partitions. Fixing the two labeled endpoints removes the duplication generated by exchanging cell names. Every such partition separates the sole conflict edge and is an immediate refinement obtained by splitting C ; hence it is minimally restorative. $\square$

For $q = 4$, [proposition 8.2](#) gives the four partitions in [proposition 8.1](#). The result shows that restoration and minimality direct inquiry but generally do not select its output.

9 A structured but fallible representation-producing correspondence

The partition lattice treats every binary refinement as equally immediate. To distinguish a one-coordinate factor from a two-coordinate interaction or exception, the representation-producing process requires some comparison or descriptive structure.

Define the analyst-level coordinate functions

$$\rho(x_{rc}) = r, \quad \chi(x_{rc}) = c.$$

These functions describe analyst-level capacities of the non-deliberative generative system to compare retained particulars. They are not initially elements of the firms' subjective event algebra or information possessed for deliberation. Before a representation is produced, the firm cannot name a row or column predicate or optimize over the corresponding partitions.

For any binary partition $\mathcal{P}$, let $b_{\mathcal{P}} : X \rightarrow \{0,1\}$ be an indicator of either one of its cells. Coordinate ρ is essential to $b_{\mathcal{P}}$ if changing the row while holding the column fixed changes the indicator for some column. Define essentiality of χ analogously. Complementing $b_{\mathcal{P}}$ does not alter its essential coordinates.

Definition 9.1 (Coordinate complexity). *The coordinate complexity $\kappa(\mathcal{P})$ of a binary partition is the number of coordinates in $\{\rho, \chi\}$ essential to an indicator generating that partition.*

It follows that

$$\kappa(\mathcal{P}^r) = \kappa(\mathcal{P}^c) = 1, \tag{19}$$

whereas

$$\kappa(\mathcal{P}^{\times}) = \kappa(\mathcal{P}^{jk}) = 2 \quad \text{for every } j, k \in \{0,1\}. \tag{20}$$

Rows and columns are one-factor distinctions. The diagonal partition is an interaction, while a singleton exception requires a conjunction of the two coordinates.

Definition 9.2 (Complexity-filtered restorative correspondence). *For any $(\mathcal{P}, D)$ whose minimally restorative refinements are binary, define*

$$\mathcal{I}(\mathcal{P}, D) = \arg \min_{\mathcal{P}' \in \mathcal{R}(\mathcal{P}, D)} \kappa(\mathcal{P}'). \tag{21}$$

Corollary 9.3 (Factor-based candidates). *For the discovery data,*

$$\mathcal{I}(\mathcal{P}^0, \overline{D}) = \{\mathcal{P}^r, \mathcal{P}^c\}.$$

Proof. Apply equations (19) and (20) to the four partitions in equation (18). □

The correspondence is not the firm's subjective answer menu and κ is not a cost the unaware firm consciously minimizes. It is the analyst's model of which conceptual products can be generated from the experienced contrast by a particular representational capacity. The imposed grammar is substantive and contestable: it favors simple factor explanations, even in a world where a conjunctive exception is objectively correct. Alternative inquiry technologies can be studied by varying the available grammar and its complexity ordering.

Definition 9.4 (Representation-producing technology). *A deterministic representation-producing technology for firm i is an analyst-level function*

$$\phi_i : \Pi(X) \times H \longrightarrow \Pi(X).$$

The output $\phi_i(\mathcal{P}, h) = \mathcal{P}$ means that no new representation is produced. A strict output $\mathcal{P} \prec \phi_i(\mathcal{P}, h)$ is a successful production event. The technology is restorative if every strict output lies in $\mathcal{R}(\mathcal{P}, D(h))$. When $\mathcal{I}(\mathcal{P}, D(h))$ is defined, it is factor-seeking if every strict output lies in $\mathcal{I}(\mathcal{P}, D(h))$.

The technology is not controlled through subjective optimization. Its input h is the analyst's representation of the retained particulars on which the non-deliberative generative system can operate. Its output, and even the set of outputs it could have produced, is not a pre-insight subjective menu.

Assumption 9.5 (Dissonance-triggered production). *If $\widehat{M}(\mathcal{P}, D(h)) \neq \emptyset$, then $\phi_i(\mathcal{P}, h) = \mathcal{P}$. If the active set is empty, the technology may fail, again producing $\mathcal{P}$, or successfully produce a minimally restorative refinement. The main factor-seeking specification further restricts a successful output to $\mathcal{I}(\mathcal{P}, D(h))$. If production fails while the active set is empty, the firm carries forward its previously committed policy until a successful production event occurs. All results that condition on an acquired refinement condition on such success.*

The function ϕ_i has no direct objective-law argument. Two causal worlds that generate the same retained history and current awareness therefore produce the same output. Agent-specific prior understanding, analogy, and organizational routines may be represented by differences across the functions ϕ_i , but not by allowing a function to inspect unobserved values of the objective law.

10 Nonidentification and the cost of robustness

The focal truth is the row law $d^r \equiv d$, where $d^r(x_{rc}) = r$. For comparison, define the column law

$$d^c(x_{rc}) = c. \quad (22)$$

Both causal worlds agree on the discovery states:

$$d^r(x_{00}) = d^c(x_{00}) = 0, \quad d^r(x_{11}) = d^c(x_{11}) = 1.$$

Definition 10.1 (Truth-blind inquiry-and-formulation rule). *A single-valued inquiry-and-formulation rule returns, conditional on successful representation production, one pair $(\mathcal{P}', m')$ satisfying*

$$\mathcal{P}' = \phi_i(\mathcal{P}, h), \quad m' \in \widehat{M}(\mathcal{P}', D(h)).$$

It is truth-blind if two objective causal laws that generate the same current awareness partition and retained history induce the same pair. The rule is correct under objective law d' if $m' = d'$. It may depend on experience, retained particulars, the comparison grammar, and agent-specific background capacities, but not on unobserved values of the objective law.

Theorem 10.2 (Observational nonidentification). *No deterministic truth-blind inquiry-and-formulation rule can produce the correct causal formulation after $\overline{D}$ in both the row world d^r and the column world d^c .*

Proof. The current partition and retained history are identical in the two worlds, so a deterministic truth-blind rule must produce the same formulation in both. But $d^r \neq d^c$. The same formulated theory cannot equal both objective laws. $\square$

This result does not depend on the coordinate-complexity criterion. It follows from observational equivalence. The 2×2 geometry provides an additional symmetry interpretation. Let

$$\pi(x_{rc}) = x_{cr}$$

exchange the coordinates. The transformation fixes x_{00} , x_{11} , $\mathcal{P}^0$, and $\overline{D}$, while exchanging $\mathcal{P}^r$ and $\mathcal{P}^c$. For a partition selector s , coordinate equivariance means

$$s(\pi\mathcal{P}, \pi h) = \pi s(\mathcal{P}, h).$$

Here $\pi\mathcal{P}$ and πh apply the state relabeling to every partition cell and every customer-state entry in the retained history. Therefore no deterministic coordinate-equivariant selector can choose one of the two elements of $\mathcal{I}(\mathcal{P}^0, \overline{D})$ without an additional asymmetry in evidence or background accessibility. The complexity filter, rather than symmetry alone, is what excludes the invariant singleton- exception candidates.

Definition 10.3 (Causal sufficiency). *Partition $\mathcal{P}$ is sufficient for a causal law d' if*

$$d' \in M(\mathcal{P}),$$

or, equivalently, if d' is constant on every cell of $\mathcal{P}$.

Corollary 10.4 (Worst-case success of minimal inquiry). *For comparison with the deterministic baseline, suppose an analyst-level truth-blind technology randomizes over $\mathcal{R}(\mathcal{P}^0, \overline{D})$, and call it successful in a causal world when its selected partition is sufficient to express that world's objective law. Randomized truth blindness requires observationally equivalent causal worlds to induce the same distribution over outputs. Its worst-case probability of success across $\{d^r, d^c\}$ is at most $1/2$. The bound is attained by assigning equal probability to $\mathcal{P}^r$ and $\mathcal{P}^c$.*

Proof. Among the four minimally restorative partitions, only $\mathcal{P}^r$ is sufficient for d^r and only $\mathcal{P}^c$ is sufficient for d^c . If their selection probabilities are p_r and p_c , then $p_r + p_c \leq 1$, so $\min\{p_r, p_c\} \leq 1/2$. $\square$

Proposition 10.5 (Robust sufficiency requires complete awareness). *If a partition is sufficient for both d^r and d^c , then it is the singleton partition*

$$\mathcal{P}^{\text{full}} = \{\{x_{00}\}, \{x_{01}\}, \{x_{10}\}, \{x_{11}\}\}. \quad (23)$$

Proof. Sufficiency for d^r requires every cell to lie within a row. Sufficiency for d^c requires every cell to lie within a column. Each row-column intersection contains exactly one state. Hence every cell is a singleton. $\square$

Proposition 10.5 concerns representation, not identification. Complete awareness makes both observationally equivalent causal laws expressible, but $\overline{D}$ alone still does not determine which one is true. Thus exhaustive awareness is necessary for a representation guaranteed to accommodate both worlds, but is not sufficient for knowledge. A non-exhaustive technology avoids brute-force bookkeeping only by accepting fallibility.

11 Formulation, representation-relative coherence, and judgment

Successful representation production first supplies a new partition $\mathcal{P}'$. A causal formulation is then a pair

$$(\mathcal{P}', m'), \quad m' \in \widehat{M}(\mathcal{P}', \overline{D}).$$

In the focal example, every minimally restorative partition in [equation \(18\)](#) has a unique active theory, so representation and formulation occur in immediate succession. They remain conceptually distinct: the partition supplies a distinction; the theory states the causal relation organized by that distinction.

After acquiring $\mathcal{P}'$, firm i forms a bridge prior

$$\mu_i^+ \in \Delta(M(\mathcal{P}')). \quad (24)$$

Ordinary Bayesian updating does not derive μ_i^+ from μ_i^0 , because the domains differ. For the deterministic example, assume only that the bridge prior has full support. Define the representation-relative coherence distribution

$$\bar{\mu}_i(m \mid \mathcal{P}', \bar{D}) = \frac{\mu_i^+(m) \mathbf{1}\{m \in \hat{M}(\mathcal{P}', \bar{D})\}}{\sum_{m' \in M(\mathcal{P}')} \mu_i^+(m') \mathbf{1}\{m' \in \hat{M}(\mathcal{P}', \bar{D})\}}. \quad (25)$$

This is a retrospective renormalization on the newly produced model space, not a posterior generated from one temporally prior probability model over the entire representation-selection process.

Proposition 11.1 (Representation-relative certainty under a false formulation). *Suppose inquiry produces $\mathcal{P}^c$ and μ_i^+ has full support on $M(\mathcal{P}^c)$. The coherence distribution assigns probability one to*

$$m^c = (0, 1, 0, 1).$$

Nevertheless, under the row truth,

$$m^c(x_{01}) \neq d^r(x_{01}), \quad m^c(x_{10}) \neq d^r(x_{10}).$$

Proof. Under $\mathcal{P}^c$, cell constancy and $x_{00} \mapsto 0$ require value zero on the first column. Cell constancy and $x_{11} \mapsto 1$ require value one on the second. Hence m^c is the unique model in $\hat{M}(\mathcal{P}^c, \bar{D})$, and full support gives it $\bar{\mu}_i$ -probability one. Direct evaluation at the off-diagonal states establishes both errors. $\square$

The distribution in [proposition 11.1](#) is Bayesian-coherent conditional on the generated representation. It should not be treated as independent confirmation: the same evidence directed representation production and fitted the resulting theory. To implement reflective judgment cleanly in this model, evidential assessment must use fresh evidence or a likelihood that explicitly accounts for the representation-selection process. The remainder of this section makes that accounting exact.

Definition 11.2 (Reflective assessment problem). *A reflective assessment problem is a pair $(\mathcal{D}, \nu)$, where $\mathcal{D}$ is a finite set of candidate objective laws $d' : X \rightarrow A$ and $\nu \in \Delta(\mathcal{D})$ is a full-support reflective prior. The assessment asks how much probability the total evidence—the retained history together with the record of what inquiry produced—places on each candidate law.*

Definition 11.3 (Selection-adjusted warrant and the warrant gap). *Fix a history h and a successful production event yielding the formulation $(\mathcal{P}', m')$. The selection-adjusted warrant of the formulation is*

$$W(m' \mid h) = \nu(\{d \in \mathcal{D} : d = m'\} \mid h, \phi_i(\mathcal{P}, h) = \mathcal{P}'), \quad (26)$$

the posterior probability that the objective law coincides with the formulated theory, conditional on both the retained history and the production event. The warrant gap is

$$G(m' \mid h) = \bar{\mu}_i(m' \mid \mathcal{P}', D(h)) - W(m' \mid h). \quad (27)$$

Lemma 11.4 (Guaranteed fit). *Conditional on a successful production event, the produced partition lies in $\mathcal{R}(\mathcal{P}, D(h))$, so its active set is nonempty and every active theory agrees with all revealed labels. The event that the formulation fits the discovery data therefore has probability one under every candidate law that generates h , and retrospective fit carries no discriminating information.*

Proof. Immediate from [definition 9.4](#) and [equation \(10\)](#): a restorative output is defined by the nonemptiness of its active set, and membership of the active set is defined by agreement with $D(h)$. Neither condition involves the objective law except through $D(h)$ itself. $\square$

Proposition 11.5 (Prior preservation under truth-blind production). *Suppose the firms' committed policies are fixed, the technology is truth-blind (deterministic or randomized), and each candidate law consistent with the revealed labels generates the same realized history h . Then conditioning on h and on the production event excludes the candidates inconsistent with the revealed labels and preserves the reflective prior odds among the rest. For the discovery history with $\mathcal{D} = \{d^r, d^c\}$ and symmetric ν , both candidates survive; production of either factor-based refinement yields a coherence probability of one on its unique active theory (proposition 11.1); and*

$$W(m^r | h) = W(m^c | h) = \frac{1}{2}, \quad G(m^r | h) = G(m^c | h) = \frac{1}{2}.$$

Proof. The reflective posterior on $d \in \mathcal{D}$ is proportional to the product of three factors: the prior $\nu(d)$; the likelihood of the realized history, which in the deterministic environment is an indicator equal to one exactly when d agrees with every label in $D(h)$, since a disagreeing law would have produced different payoffs at some differentiated episode; and the probability of the production event given (d, h) , which by truth blindness depends on $(\mathcal{P}, h)$ alone and is therefore constant across $\mathcal{D}$ and cancels. This establishes the general claim. For the discovery history, d^r and d^c agree with both labels in $\bar{D}$ (theorem 10.2), so the symmetric prior is preserved: $\nu(d^c | \cdot) = \nu(d^r | \cdot) = 1/2$. Since $m^c = d^c \neq d^r$, the warrant of m^c is one-half, and symmetrically for m^r . Subtracting from the unit coherence probability gives the gap. $\square$

Corollary 11.6 (Self-audit requires complete awareness). *The reflective assessment $(\mathcal{D}, \nu)$ is expressible by firm i only if every candidate law in $\mathcal{D}$ is a theory expressible under $\mathcal{P}_i$, that is, only if $\mathcal{P}_i$ is sufficient for every $d \in \mathcal{D}$. For $\mathcal{D} = \{d^r, d^c\}$, proposition 10.5 implies $\mathcal{P}_i = \mathcal{P}^{\text{full}}$. A firm holding either minimally restorative factor candidate therefore cannot formulate the comparison that measures its own warrant gap: the overstatement identified in proposition 11.1 is invisible from within the representation that produces it. Short of complete awareness, warrant must be supplied externally—by fresh diagnostic evidence, an analyst, a differently aware rival, or a future self with expanded awareness.*

Proof. Holding ν over $\mathcal{D}$ requires entertaining each candidate law as a hypothesis about customer response, which requires each law to be cell constant under $\mathcal{P}_i$ (section 4); this is causal sufficiency for each $d \in \mathcal{D}$. Sufficiency for both d^r and d^c forces the singleton partition by proposition 10.5. $\square$

Remark 11.7 (Double use, made exact). *The coherence distribution treats restored fit as confirmation; lemma 11.4 shows the fit was guaranteed by the restoration property, and proposition 11.5 shows the production event carries no additional information when the technology is truth-blind. The warrant gap of one-half is therefore an exact measure of the confidence overstatement produced by using the discovery experience twice—once to direct representation production and once to fit the resulting theory. Truth blindness is the polar case in which the production event is completely uninformative. If production were correlated with the objective law—for instance, social transmission from an informed source—the event would itself be evidence, and the selection adjustment could raise as well as lower confidence.*

Remark 11.8 (Unwarranted certainty is not a symptom of error). *The gap afflicts the correct formulation equally: the firm that acquires $\mathcal{P}^r$ holds $\bar{\mu} = 1$ on $m^r = d^r$, yet its warrant at the discovery stage is also one-half. Early certainty is unwarranted even when it happens to be true. What separates the two firms is not their evidence or their coherence but the objective law, which neither observes; validation by fresh differentiated evidence is what converts a fortunate formulation into warranted knowledge.*

The row and column formulations make opposing off-diagonal predictions:

Formulation	x_{00}	x_{01}	x_{10}	x_{11}
Row m^r	0	0	1	1
Column m^c	0	1	0	1

Either off-diagonal state can therefore discriminate between them, provided strategic actions reveal its objective label.

Proposition 11.9 (Diagnostic closure of the warrant gap). *Append additional episodes to the discovery history and maintain the reflective assessment of [proposition 11.5](#).*

- (i) *A single differentiated episode at an off-diagonal state closes the gap. If the revealed label agrees with the acquired formulation, W rises to one and $G = 0$. If it disagrees, the active set under the acquired partition empties, coherence collapses with renewed dissonance, and the warrant of the surviving candidate rises to one.*
- (ii) *Pooled episodes, at any states and in any number, change neither the coherence distribution nor the selection-adjusted posterior: the warrant gap is invariant under unbounded pooled experience.*

Proof. (i) At x_{01} the candidates predict opposite labels (d^r assigns 0 and d^c assigns 1), and likewise at x_{10} . A revealed label therefore has likelihood one under one candidate and zero under the other, making the reflective posterior degenerate. If the label agrees with the acquired formulation, the active set is unchanged, $\bar{\mu}$ remains one, and $G = 1 - 1 = 0$. If it disagrees, the new label conflicts with a previously revealed diagonal label inside one cell of the acquired partition, the active set empties, and [assumption 9.5](#) governs renewed inquiry. (ii) By [lemma 6.2](#), a pooled episode has the same payoff vector under every candidate law, so its likelihood is one under both candidates and the reflective posterior is unchanged; the active set is also unchanged, so $\bar{\mu}$ is unchanged. $\square$

Proposition 11.10 (Resolution under one additional diagnostic label). *Suppose representation production initially fails after $\overline{D}$, so awareness remains $\mathcal{P}^0$ and the previously differentiated policy profile is carried forward for one additional diagnostic episode. Let the row law be true.*

- (a) *If differentiated actions reveal $x_{01} \mapsto 0$, the minimally restorative refinements are $\mathcal{P}^r$ and $\mathcal{P}^{11}$, and the unique minimum-complexity refinement is $\mathcal{P}^r$.*
- (b) *If differentiated actions reveal $x_{10} \mapsto 1$, the minimally restorative refinements are $\mathcal{P}^r$ and $\mathcal{P}^{00}$, and the unique minimum-complexity refinement is $\mathcal{P}^r$.*

Proof. For part (a), the two zero-labeled states must lie apart from x_{11} . In a minimal binary restoration, the unobserved state x_{10} may join either side, giving the row partition and the partition isolating x_{11} . Their complexities are one and two. Part (b) is symmetric. $\square$

The proposition is a continued-commitment comparative exercise rather than a claim that undefined post-dissonance beliefs generate an equilibrium policy. It is conditional on obtaining a fresh diagnostic observation. An unaware firm cannot simply be assumed to target “the off-diagonal state” as an experiment before it can formulate that distinction. A theory of inquiry must explain whether the observation arrives through ordinary experience, comparison of retained cases, social transmission, or an exploration technology available without prior semantic specification.

12 Cumulative awareness and path dependence

Assumption 12.1 (Cumulative awareness). *Once a distinction becomes available, later awareness can refine but does not coarsen the acquired partition. Rejecting a causal theory need not make the distinction on which it was formulated unthinkable again.*

This is an explicit economic assumption, not a logical implication of unawareness. It permits causal theory to be rejected while the associated conceptual distinction remains available.

Proposition 12.2 (Path-dependent terminal awareness). *Assume the row law, cumulative awareness, successful minimally restorative revision after every dissonance, and eventual differentiated observations at both off-diagonal states.*

- (i) *A firm that initially acquires $\mathcal{P}^r$ holds the correct formulation $m^r = d^r$ and need not refine further.*
- (ii) *A firm that initially acquires $\mathcal{P}^c$ must eventually reach $\mathcal{P}^{\text{full}}$.*

The two firms can therefore converge to the same correct causal theory and policy while retaining different awareness partitions.

Proof. Under $\mathcal{P}^r$, every subsequently revealed label agrees with m^r , so the active set never becomes empty.

Under $\mathcal{P}^c$, x_{00} and x_{10} initially share one cell, while x_{01} and x_{11} share the other. A revealed label at x_{01} conflicts with the previous label at x_{11} and forces the second column to split. A revealed label at x_{10} conflicts with the previous label at x_{00} and forces the first column to split. Reversing the order of the off-diagonal observations reverses the order of these two splits. Cumulative awareness rules out discarding the column distinction and replacing it with the incomparable row partition. The terminal partition is therefore the coarsest common refinement

$$\mathcal{P}^c \vee \mathcal{P}^r = \mathcal{P}^{\text{full}}.$$

Once all four labels have been revealed, the unique active causal function is d^r under either awareness path. $\square$

Remark 12.3 (Alternative false paths). *If the complexity filter is not imposed, an initial singleton formulation also leaves a permanent representational footprint. Under complete diagnostic coverage, $\mathcal{P}^{00}$ leads to*

$$\{\{x_{00}\}, \{x_{01}\}, \{x_{10}, x_{11}\}\},$$

whereas $\mathcal{P}^{11}$ leads to

$$\{\{x_{00}, x_{01}\}, \{x_{10}\}, \{x_{11}\}\}.$$

Both are sufficient for the row law but finer than the minimally sufficient row partition.

Proposition 12.4 (Diagnostic-coverage benchmark). *Let X be finite and $d : X \rightarrow \{0, 1\}$ deterministic. Suppose awareness only changes after dissonance through strict refinements; after every dissonance, revision continues until a partition with a nonempty active set is produced; and every $x \in X$ is eventually observed under differentiated actions. Let the terminal partition be the partition in place after the last revealing observation and any revision it requires. Then the terminal partition exists, every terminal awareness cell is homogeneous under d , and the terminal active model set is $\{d\}$.*

Proof. Every successful awareness change strictly increases the number of cells, so finiteness of X permits only finitely many such changes. Hence a terminal partition exists after the final revealing observation and required revision. Differentiated observation reveals $d(x)$ for every state. An active theory must consequently satisfy $m(x) = d(x)$ for all $x \in X$, so it equals d . No terminal awareness cell can contain two states with different objective labels, because cell constancy would then make the active set empty. $\square$

Proposition 12.4 is a benchmark rather than a headline result. Its strong coverage condition does most of the work. The economically substantive issue is whether endogenous strategy supplies or suppresses differentiated observations.

13 Strategic diagnosticity and self-confirming awareness

The policies analyzed in this section are provisional commitments based on representation-relative beliefs. They may themselves generate validation evidence; they are not assumed to follow a completed reflective judgment.

Once a partition is acquired prospectively, a firm with a degenerate belief on m follows the policy

$$\sigma_i(C) = m(C)$$

at every cell labeled zero or one. By **lemma 5.2**, this policy is a subjectively strict best response against either rival action.

Definition 13.1 (Self-confirming awareness trap). *A profile of awareness partitions, beliefs, and policies forms a self-confirming awareness trap if*

- (i) *every firm's policy is optimal under its belief;*
- (ii) *all outcomes generated by the policy profile are consistent with each firm's active causal theory, so no dissonance triggers further inquiry; and*
- (iii) *there exist a firm i with active theory m_i and a state x such that $m_i(x) \neq d(x)$, but the on-path policy profile pools at x and hence does not reveal $d(x)$.*

Theorem 13.2 (Shared false formulation suppresses its own falsification). *Suppose the row law is true, both firms acquire $\mathcal{P}^c$, both become certain of $m^c = (0, 1, 0, 1)$ after the discovery batch, and every customer state subsequently recurs. Suppose **assumption 9.5** governs later representation production and the firms recommit their optimal policies in every generation. The induced policy profile is a self-confirming awareness trap. The false formulation is never falsified, even though it is wrong at both off-diagonal states.*

Proof. Under certainty of m^c , **lemma 5.2** makes the column policy

$$\sigma_i(x_{00}, x_{01}, x_{10}, x_{11}) = (0, 1, 0, 1)$$

subjectively strictly optimal for each firm. The firms therefore choose the same action at every state. By **lemma 6.2**, every subsequent payoff vector is $(1/2, 1/2)$ and supplies no causal label. The pooled payoffs are exactly those predicted by m^c under the on-path action profile, so the active set never becomes empty. Nevertheless, $m^c(x_{01}) = 1 \neq 0 = d^r(x_{01})$ and $m^c(x_{10}) = 0 \neq 1 = d^r(x_{10})$. $\square$

The trap relies on the market-splitting payoff in **equation (2)**: pooled firms receive one-half even when both offer the objectively inferior architecture. Introducing an outside option or demand losses could make an incorrect pooled architecture directly informative and would alter the result.

Corollary 13.3 (Permanent warrant gap at the trap). *In the environment of theorem 13.2, every post-discovery episode is pooled, so by proposition 11.9(ii) the warrant gap remains $G = \frac{1}{2}$ in every generation. The firms accumulate unbounded experience, remain representation-relatively certain of m^c , and are never more than half-warranted; the missing half is exactly the reflective weight on the causal world their shared partition cannot express.*

Proof. Immediate from theorem 13.2 and proposition 11.9(ii). □

Corollary 13.4 (Theory heterogeneity creates diagnostic competition). *Suppose the row law is true, firm 1 is certain of m^r , and firm 2 is certain of m^c . Their policies differ at x_{01} and x_{10} , revealing the objective labels and falsifying m^c . If all four states are served once with equal weight in one committed validation batch before either firm can revise, firm 1 earns average payoff $3/4$ and firm 2 earns $1/4$.*

Proof. The policies are

$$\sigma_1 = (0, 0, 1, 1), \quad \sigma_2 = (0, 1, 0, 1).$$

They pool at x_{00} and x_{11} , giving each firm one-half. At x_{01} and x_{10} , firm 1 offers the objectively preferred architecture and receives one. Thus its equally weighted payoff is

$$\frac{1}{4} \left(\frac{1}{2} + 1 + 1 + \frac{1}{2} \right) = \frac{3}{4},$$

with the complementary payoff accruing to firm 2. Both differentiated outcomes reveal the labels on which m^c is wrong. □

The theorem and corollary identify an endogenous epistemic effect of strategic interaction. Shared theories align actions and can suppress causal evidence. Different theories create action differentiation, which both reallocates value capture and generates the evidence needed for judgment.

14 The economics of inquiry: costly activation and delabeled leverage

In the preceding sections, representation production is triggered by dissonance and costs nothing, as if activation were free and automatic. A theory of directed inquiry requires more: whether to prosecute a question must itself be a decision. The obstacle is that the natural criterion for that decision—the expected value of the answer—is doubly unavailable at exactly the moment the question exists. The Bayesian posterior is undefined (lemma 6.3), and the candidate answers are not objects of pre-insight deliberation (definition 9.4). This section shows that the laboratory nevertheless supports an honest valuation: a statistic computed entirely from retained episodes and revealed payoffs that is invariant across every representation successful inquiry could produce.

Assumption 14.1 (Costly activation). *At the close of a generation whose history exhibits causal dissonance, firm i elects an inquiry effort $\varepsilon_i \in \{0, 1\}$ at cost $\gamma_i \varepsilon_i$, with $\gamma_i > 0$ in payoff units. The technology ϕ_i of definition 9.4 operates only if $\varepsilon_i = 1$ and then succeeds or fails as in assumption 9.5. If $\varepsilon_i = 0$ or production fails, the firm carries forward its previously committed policy. Firm i holds a delabeled success weight $\theta_i \in (0, 1]$: a subjective probability that an elected inquiry episode produces a usable refinement. The weight is defined on the event of success alone; it is not a distribution over the identities of possible outputs.*

The success weight is awareness of unawareness in delabeled form. The firm can believe that inquiry may yield a usable distinction without being able to describe any candidate distinction. The earlier sections are the special case $\gamma_i = 0$, in which election is always subjectively justified and activation is effectively automatic.

Definition 14.2 (Revealed labels and the disagreement set). A retained episode s in $e_i(h; \mathcal{P}_i)$ is labeled if $a_{1s} \neq a_{2s}$; its label a_s^* is the architecture of the firm that received payoff one. Let σ_i denote firm i 's currently committed policy. The disagreement set is

$$L_i(h) = \{s : s \text{ labeled and } \sigma_i(C_{\mathcal{P}_i}(x_s)) \neq a_s^*\},$$

the labeled episodes at whose recurrence the committed policy prescribes the architecture that lost. The retained-record leverage is

$$B_i(h) = \frac{1}{2} |L_i(h)|. \quad (28)$$

Both objects are deliberative-level computations. A token's awareness cell, the actions taken, and the payoffs received are all recorded in $e_i(h; \mathcal{P}_i)$, so identifying the winning architecture requires no predicate the firm lacks. For a firm with coarse awareness and a constant committed policy, $L_i(h)$ is exactly the set of differentiated episodes the firm lost.

Assumption 14.3 (Subjective recurrence and pinned valuation). In evaluating inquiry, firm i proceeds as if the next generation's batch reproduces its retained record token for token, and it evaluates one generation ahead. It values a prospective refinement by its pinned gain: the component of the anticipated payoff change that is common to every restorative refinement. The recurrence premise is a belief, not foresight; the analyst does not require the future batch to satisfy it, and a miscalibrated premise can produce both under- and over-investment in inquiry. A longer subjective horizon scales the valuation by the number of anticipated recurrences and changes nothing qualitative.

Lemma 14.4 (Unit leverage of a disagreeing episode). Consider the recurrence of a retained labeled episode s and compare two policies for firm i : the carried-forward committed policy, and a policy identical to it except that it implements the label a_s^* at that recurrence. Against every rival action, pure or mixed:

- (i) if $s \in L_i(h)$, the implementing policy gains exactly one-half;
- (ii) if $s \notin L_i(h)$, the two policies coincide at the recurrence and the gain is zero.

Proof. For (i), write a for the committed action at the recurrence and $a^* = a_s^* \neq a$ for the label; by lemma 6.2, $a^* = d(x_s)$. If the rival plays a^* , the committed policy is differentiated and loses, yielding zero, while the implementing policy pools, yielding one-half. If the rival plays a , the committed policy pools, yielding one-half, while the implementing policy is differentiated and wins, yielding one. In both cases the difference is one-half, and linearity extends the conclusion to mixtures. For (ii), a labeled episode outside $L_i(h)$ satisfies $\sigma_i(C_{\mathcal{P}_i}(x_s)) = a_s^*$, so implementation changes nothing. $\square$

The gain in part (i) is independent of the rival's behavior, so no conjecture about the rival is needed to price the disagreement set. The structural reason is the same one driving lemma 5.2: with two architectures and payoffs summing to one, switching from the losing to the winning architecture at a state moves the firm up by exactly one-half against either rival action.

Lemma 14.5 (Answer invariance). Let the current partition exhibit causal dissonance at $D(h)$, let $\mathcal{P}' \in \mathcal{R}(\mathcal{P}_i, D(h))$ be any minimally restorative refinement, and let $m' \in \hat{M}(\mathcal{P}', D(h))$ be held with certainty. Then the induced subjectively optimal policy implements a_s^* at the recurrence of every labeled episode s , and the anticipated pinned gain over the carried-forward policy equals $B_i(h)$, identically across all elements of $\mathcal{R}(\mathcal{P}_i, D(h))$ and their active theories. At unlabeled recurrences the induced policy may also depart from the carried-forward policy; the value of that departure depends on unrevealed values of the objective law and is not represented in $B_i(h)$.

Proof. By [proposition 7.2](#), every cell of $\mathcal{P}'$ is conflict-free, so by [equation \(10\)](#) every active theory satisfies $m'(x_s) = a_s^*$ at each labeled episode. Under certainty of m' , [lemma 5.2](#) makes the policy $\sigma_i(C) = m'(C)$ subjectively strictly optimal at every cell labeled zero or one, and [assumption 4.2](#) licenses its prospective implementation. Summing [lemma 14.4](#) over labeled recurrences gives a total pinned gain of $|L_i(h)|/2 = B_i(h)$, which does not depend on which restorative refinement was acquired. $\square$

Remark 14.6 (Delabeled deliberation). *The leverage computation quantifies over retained tokens and their recorded outcomes, never over candidate representations. The firm's reasoning can be glossed: "customers like the episodes I lost will recur; whatever distinction inquiry may yield, possessing it would let me offer such customers the architecture that won." The gloss uses only token-indexicals, which retention supplies ([assumption 4.2](#)); it requires no predicate, no answer menu, and no probability over semantic alternatives. This is the model's precise sense in which a question can be priced before its answer can be thought.*

Remark 14.7 (Pinned versus extrapolated value). *The statistic B_i prices only what every answer must deliver: agreement with the revealed labels. A full formulation extrapolates beyond them—for example, the column formulation's off-diagonal prescriptions—and the extrapolated component is exactly what representation-relative coherence cannot warrant ([proposition 11.1](#)) and what reflective judgment must assess with fresh or selection-adjusted evidence. Anticipated leverage and warranted confidence therefore separate cleanly: the former is answer-invariant and computable at dissonance; the latter is answer-specific and computable only after formulation.*

Proposition 14.8 (Participation rule). *Under [assumptions 14.1](#) and [14.3](#), electing inquiry has represented value $\theta_i B_i(h) - \gamma_i$ and abstention has represented value zero, so firm i elects inquiry if and only if*

$$\theta_i B_i(h) \geq \gamma_i,$$

with indifference resolved toward inquiry. After the discovery batch $\bar{D}$, $|L_i(h)| = 1$ for both firms, so each elects inquiry if and only if $\theta_i/2 \geq \gamma_i$.

Proof. Election costs γ_i and, with subjective probability θ_i , production succeeds; by [lemma 14.5](#) the pinned gain on the recurrence record is then $B_i(h)$ whichever refinement arrives, and [assumption 14.3](#) assigns the extrapolated component no represented value at the election. On failure the carried-forward policy continues and the gain is zero. For the discovery batch, firm 1's constant policy 0 agrees with the label at x_{00} and disagrees at x_{11} , while firm 2's constant policy 1 disagrees only at x_{00} ; hence $|L_i| = 1$ and $B_i = 1/2$ for both firms. $\square$

Both firms hold identical leverage despite opposite experiences: each won one episode and lost the other. The incentive to inquire is symmetric; which firm succeeds, and on which refinement, is governed by the technologies ϕ_i , the locus of firm-specific insight capacity.

Proposition 14.9 (Action separation, not residual uncertainty). *$B_i(h) > 0$ if and only if some retained labeled episode contradicts the committed policy. In particular:*

- (i) *on the self-confirming path of [theorem 13.2](#), no episode is ever labeled, so $L_i(h) = \emptyset$ and $B_i(h) = 0$ for both firms in every generation, although each firm's formulation is false at two of the four states and the empty label record leaves the entire model space unrestricted at the analyst's level;*
- (ii) *after the discovery batch, the revealed labels reduce the minimally restorative refinements to four ([proposition 8.1](#)) and the complexity-filtered candidates to two ([corollary 9.3](#)), yet $B_i(h) = 1/2 > 0$.*

The represented value of a question therefore tracks the action separation in the retained record, not the quantity of unresolved structure.

Proof. The equivalence restates [definition 14.2](#). For (i), pooled actions label no episode by [lemma 6.2](#). For (ii), the count is computed in [proposition 14.8](#), and the reduction of candidates is given by the cited results. $\square$

Corollary 14.10 (Trap immunity to cheap inquiry). *Extend the environment of [theorem 13.2](#) with [assumption 14.1](#). The trap persists for every $\gamma_i > 0$ and every $\theta_i \leq 1$. It is doubly locked: on-path experience generates no dissonance, so the technology is inert ([assumption 9.5](#)); and $B_i(h) = 0$ in every generation, so even an infallible technology at an arbitrarily small positive cost would receive no represented justification. Escaping the trap therefore requires a source of action differentiation outside the represented calculus—exogenous experimentation, entry by a firm with a different formulation, or a payoff environment in which pooled misservice is itself informative.*

Proof. Combine [theorem 13.2](#) with [proposition 14.9\(i\)](#) and the participation rule of [proposition 14.8](#). $\square$

Corollary 14.11 (Rational cessation and the loser’s leverage). *Append the validation batch of [corollary 13.4](#) to the discovery history, with firm 1 certain of m^r and firm 2 certain of m^c .*

- (i) *Firm 1’s record contains labels at all four states, each agreeing with its committed policy: $L_1 = \emptyset$ and $B_1 = 0$. Firm 1 experiences no dissonance and, even were inquiry offered, would decline it at every positive cost. Its terminal awareness $\mathcal{P}^r$ remains incomplete ([proposition 10.5](#)), and that incompleteness is subjectively optimal rather than merely mechanically absorbing.*
- (ii) *Firm 2’s record contains the same four labels; its committed column policy disagrees at x_{01} and x_{10} , so $|L_2| = 2$ and $B_2 = 1$ —twice the discovery-stage leverage. Firm 2 experiences dissonance, elects inquiry whenever $\theta_2 \geq \gamma_2$, and, conditional on success, proceeds by [proposition 12.2](#) toward complete awareness.*

Differentiated defeat thus concentrates the incentive to inquire in the defeated firm: the same episodes that transfer value to the correctly formulated rival supply the loser with dissonance, revealed labels, and leverage. Competitive failure is, in this precise sense, an engine of awareness growth, and sustained success rationally extinguishes the question.

Proof. The labels revealed across the two batches are $x_{00} \mapsto 0$, $x_{11} \mapsto 1$ (discovery) and $x_{01} \mapsto 0$, $x_{10} \mapsto 1$ (validation). Firm 1’s policy $(0, 0, 1, 1)$ agrees with every label, so $L_1 = \emptyset$; absence of dissonance follows because $m^r \in \hat{M}(\mathcal{P}^r, D(h))$. Firm 2’s policy $(0, 1, 0, 1)$ disagrees at exactly the two off-diagonal states, so $|L_2| = 2$. Its dissonance follows because under $\mathcal{P}^c$ the label $x_{01} \mapsto 0$ conflicts with $x_{11} \mapsto 1$ in one cell and $x_{10} \mapsto 1$ conflicts with $x_{00} \mapsto 0$ in the other, emptying the active set. The election claim applies [proposition 14.8](#). $\square$

Remark 14.12 (Relation to deliberation-cost models). *The participation rule instantiates, in partition form, the deliberation logic of the C-family agents of [2]: refinement effort is expended only when its represented value exceeds its cost. There, the value of investigating an unexplored branch is computed from objective payoff bounds attached to unseen continuations. The present treatment replaces those bounds with the firm’s own revealed record, in keeping with the discipline that nothing behind the awareness frontier may carry objective payoff information into pre-insight deliberation.*

Remark 14.13 (An asymmetric partial-correction benchmark). Suppose instead that firm 2 remains exogenously fixed at architecture 1 and firm 1 initially follows the false column theory. A differentiated loss at x_{10} can lead firm 1 to the partition

$$\{\{x_{00}\}, \{x_{10}\}, \{x_{01}, x_{11}\}\}$$

and theory $(0, 1, 1, 1)$. Firm 1 then wins at x_{00} and pools elsewhere, earning $5/8$ under equal state weights. Its theory remains wrong at x_{01} , where pooling conceals the error. This calculation is useful as an illustration, but it is not an equilibrium theorem unless firm 2's continued fixed behavior and failure of inquiry are endogenized.

15 An outside option: value destruction and the knife-edge of self-confirmation

Under equation (2), total value is one in every episode: pooled firms split the market even when both offer the objectively inferior architecture, so misservice reallocates nothing and destroys nothing. Three results lean on that feature: the self-confirming trap (theorem 13.2), its immunity to cheap inquiry (corollary 14.10), and the permanent warrant gap (corollary 13.3). This section parameterizes the demand side and shows that all three are knife-edge properties of perfectly forgiving demand.

Definition 15.1 (Outside option). For $\beta \in [0, 1]$, replace equation (2) with

$$u_i^\beta(a_i, a_j, x) = \begin{cases} \frac{1}{2}, & a_i = a_j = d(x), \\ \frac{\beta}{2}, & a_i = a_j \neq d(x), \\ 1, & a_i \neq a_j \text{ and } a_i = d(x), \\ 0, & a_i \neq a_j \text{ and } a_j = d(x). \end{cases} \quad (29)$$

Payoffs sum to the value created: one whenever some firm offers $d(x)$, and β when both firms offer the other architecture, in which case the customer withholds $1 - \beta$ of the unit value. The subjective payoff predicted by theory m extends accordingly:

$$u_i^{m, \beta}(a_i, a_j, x) = \begin{cases} \frac{1}{2}, & a_i = a_j \text{ and } m(x) \in \{a_i, n\}, \\ \frac{\beta}{2}, & a_i = a_j \text{ and } m(x) \notin \{a_i, n\}, \\ 1, & a_i \neq a_j \text{ and } m(x) = a_i, \\ 0, & a_i \neq a_j \text{ and } m(x) = a_j, \\ \frac{1}{2}, & a_i \neq a_j \text{ and } m(x) = n. \end{cases} \quad (30)$$

The payoff rule, including β , is known to both firms; only the causal law is not. Every preceding section is the special case $\beta = 1$.

Lemma 15.2 (Dominance under an outside option). At an awareness cell C , the subjective expected-payoff differences between actions 0 and 1 are

$$v_i(0 | C) - v_i(1 | C) = \begin{cases} \frac{1}{2} p_i^0(C) - (1 - \frac{\beta}{2}) p_i^1(C), & \text{rival plays 0,} \\ (1 - \frac{\beta}{2}) p_i^0(C) - \frac{1}{2} p_i^1(C), & \text{rival plays 1,} \end{cases}$$

with mixtures handled by linearity. Consequently:

- (i) a firm certain of an architecture label finds the labeled action strictly dominant on that cell for every $\beta \in [0, 1]$;
- (ii) action 0 is strictly dominant on C if and only if $p_i^0(C) > (2 - \beta) p_i^1(C)$, and symmetrically for action 1; hence belief-induced differentiation ([proposition 5.4](#)) extends to $\beta < 1$ provided the prior asymmetry in [assumption 5.3](#) is strengthened to this threshold.

Proof. Against rival action 0: $v_i(0 | C) = \frac{1}{2}[p_i^0(C) + p_i^n(C)] + \frac{\beta}{2}p_i^1(C)$ and $v_i(1 | C) = p_i^1(C) + \frac{1}{2}p_i^n(C)$. Against rival action 1: $v_i(0 | C) = p_i^0(C) + \frac{1}{2}p_i^n(C)$ and $v_i(1 | C) = \frac{1}{2}[p_i^1(C) + p_i^n(C)] + \frac{\beta}{2}p_i^0(C)$. Subtracting gives the displayed differences, which reduce to [equation \(6\)](#) at $\beta = 1$. For (i), $p_i^n(C) = 1$ makes both differences equal to $\frac{1}{2}$ and $1 - \frac{\beta}{2}$, both strictly positive. For (ii), both differences are strictly positive exactly when $\frac{1}{2}p_i^0(C) > (1 - \frac{\beta}{2})p_i^1(C)$. $\square$

When $\beta < 1$ and neither threshold in part (ii) is met, no constant action is dominant and best responses acquire an anti-coordination structure: differentiating guarantees that the customer is served and value one is created, so demand-side harshness pushes balanced-belief play toward the diagnostic regime. A full equilibrium analysis of that game is deferred.

Lemma 15.3 (Demand-side revelation). *Let $\beta < 1$ and consider a pooled episode at state x with common action a . The payoff level reveals the objective label: payoff one-half implies $d(x) = a$ and excludes the theory value $1 - a$ at x , while payoff $\beta/2$ implies $d(x) = 1 - a$ and excludes both the value a and neutrality. Differentiated episodes reveal labels as in [lemma 6.2](#). At $\beta = 1$, pooled episodes reveal nothing.*

Proof. Objectively, [equation \(29\)](#) gives pooled payoff one-half if $a = d(x)$ and $\beta/2$ otherwise, and the two are distinguishable exactly when $\beta \neq 1$. Subjectively, [equation \(30\)](#) predicts one-half under $m(x) \in \{a, n\}$ and $\beta/2$ under $m(x) = 1 - a$; comparing predicted with realized payoffs yields the stated exclusions. $\square$

Remark 15.4 (Leverage and participation under an outside option). *For $\beta < 1$, [definition 14.2](#) extends by counting as labeled any pooled episode whose payoff level pins the objective label ([lemma 15.3](#)). The unit gain of implementing a disagreeing label is no longer conjecture-free: depending on the rival's action at the recurrence, it lies in $[\frac{1}{2}, 1 - \frac{\beta}{2}]$, so $B_i(h) = |L_i(h)|/2$ is a conjecture-free lower bound on the pinned gain rather than its exact value, and the participation rule of [proposition 14.8](#) errs only toward abstention. The results of [section 14](#) hold as stated at $\beta = 1$; for $\beta < 1$ the trap-immunity corollary loses its premise because, as shown next, the trap itself fails.*

Proposition 15.5 (No payoff-relevant self-confirmation for $\beta < 1$). *Let $\beta < 1$. Consider awareness partitions, degenerate beliefs (each firm certain of a single expressible theory), and committed policies recommitted each generation, satisfying conditions (i) and (ii) of [definition 13.1](#)—subjective optimality and on-path consistency—and suppose every customer state recurs. Then every state is served by its objectively preferred architecture, so value creation is maximal in every generation, and every architecture label is objectively correct: $m_i(x) \in \{0, 1\}$ implies $m_i(x) = d(x)$. Residual falsity is confined to neutral mislabeling—states labeled n whose realized service is nonetheless correct—which affects neither value creation nor value capture.*

Proof. Fix a recurring state x . First, the actions must pool at x . Suppose they differ; the losing firm observes payoff zero. If the loser's cell carries an architecture label, [lemma 15.2\(i\)](#) and subjective optimality require its action to equal that label, so its predicted payoff is one, and if its cell is labeled n its predicted payoff is one-half; either way the observation contradicts consistency. Second, suppose the pooled payoff is $\beta/2$. By [lemma 15.3](#), consistency requires $m_i(x) = 1 - a$ for both

firms; lemma 15.2(i) then prescribes action $1 - a$, contradicting the pooled action a . Hence the pooled payoff is one-half, so $d(x) = a$: the state is correctly served, and consistency admits only $m_i(x) \in \{a, n\}$, so any architecture label equals $d(x)$. $\square$

Corollary 15.6 (Knife-edge of the epistemic pathologies). *The column profile of theorem 13.2 is a self-confirming awareness trap if and only if $\beta = 1$. For $\beta < 1$, in the first generation containing an off-diagonal state:*

- (i) *both firms receive $\beta/2$ where their shared theory predicts one-half; the pinned label conflicts with a previously revealed diagonal label inside a cell of $\mathcal{P}^c$, the active set empties, and assumption 9.5 governs renewed inquiry;*
- (ii) *the pinned label disagrees with the committed column policy, so the extended disagreement set is nonempty, $B_i > 0$, and the zero-leverage lock of corollary 14.10 fails;*
- (iii) *the observed payoff has likelihood one under d^r and zero under d^c , so the reflective posterior is degenerate and the permanent warrant gap of corollary 13.3 fails.*

The destroyed value $1 - \beta$ at each misserved episode is itself the corrective signal: what demand withholds from the firms, their epistemics receives as evidence.

Proof. The $\beta = 1$ direction is theorem 13.2. Let $\beta < 1$ and let the off-diagonal state be x_{01} (the case x_{10} is symmetric). The column policies pool on action 1 there while $d^r(x_{01}) = 0$, so the realized payoff is $\beta/2$, whereas $m^c(x_{01}) = 1$ predicts one-half: consistency fails, so the profile is not a trap. By lemma 15.3 the label $x_{01} \mapsto 0$ is pinned; it conflicts with the discovery label $x_{11} \mapsto 1$ in the cell $\{x_{01}, x_{11}\}$ of $\mathcal{P}^c$, so the active set empties (proposition 7.2). The committed policy plays 1 at x_{01} , so the episode enters the extended disagreement set and $B_i \geq \frac{1}{2}$. Finally, under d^c the pooled action at x_{01} is correct and the predicted payoff is one-half, so the realized $\beta/2$ has likelihood zero, while under d^r it has likelihood one; the reflective posterior concentrates on d^r . $\square$

Remark 15.7 (Two sources of diagnosticity). *At $\beta = 1$, evidence arises only from strategic differentiation: theory heterogeneity across firms is the sole corrective force, and consensus on a false formulation is absorbing (theorem 13.2 and corollaries 13.3 and 14.10). For $\beta < 1$, the demand side supplies evidence precisely when the strategy side pools, and proposition 15.5 shows that payoff-relevant collective error cannot persist. The comparative interpretation is direct: markets with forgiving demand—captive customers, absent substitutes, buyers who do not bear the cost of misservice—require competitive variety in beliefs to correct false collective theories, whereas markets with strong outside options discipline theories even under consensus. For the general dynamic model, the parameter locates two learning regimes: $\beta < 1$ supplies an evidence stream independent of strategic heterogeneity, while $\beta = 1$ makes all learning contingent on it.*

16 Cognitive interpretation

The model's relationship to Lonergan's account of inquiry is interpretive, not algorithmic. Lonergan does not supply a partition lattice, a coordinate-complexity criterion, or a Bayesian rule. His distinctions instead discipline which formal objects should not be conflated. The relevant distinctions among insight, formulation, and reflective judgment are developed in Lonergan [1, chs. 1 and 10]; the question-insight-formulation-validation sequence is also developed in Ryall and Wilkins [3].

Cognitive operation	Formal counterpart	Qualification
Experience	Customer episodes, actions, and outcomes	Particular experience does not itself supply a causal predicate.
Question for intelligence	In this baseline, causal dissonance makes salient the question of what explains the reversal	The question specifies the defect and requirements of an answer, not a known answer menu.
Heuristic anticipation	Restoration, minimality, and projectibility	These are the model's analogue of heuristic anticipation: they constrain what an answer must accomplish without providing its semantic content.
Direct insight	No direct mathematical counterpart	The model represents only the ensuing representational and formulational products, not the conscious occurrence.
Representation production	New partition $\mathcal{P}'$	A previously unavailable distinction becomes usable in theory and action.
Formulation	Pair $(\mathcal{P}', m')$	A distinction alone is not a causal explanation; a stable relation must be articulated on it.
Question for reflection	Whether m' holds beyond the generating cases	Retrospective fit is not independent confirmation.
Judgment	Assessment of whether the formulation is adequately warranted	Fresh diagnostic evidence or a selection-adjusted likelihood supplies the model's implementation.
Decision	Policy measurable with respect to $\mathcal{P}'$	Strategic action determines which future experience will be diagnostic; provisional action may occur before judgment is complete.

In this interpretation, the formal sequence associated with a successful insight is not exhausted by partition production: the available distinction must subsequently be expressed in a projectible causal formulation. Restoration is weaker than truth, so the candidate formulation remains subject to reflective judgment. Agent-specific prior understanding, analogy, habits of comparison, and organizational routines can eventually be modeled as differences in heuristic accessibility or in the complexity ordering, but they must not become a subjective menu over semantically named answers before those answers are thinkable.

In Lonergan's account the desire to know is unrestricted, but its deliberate prosecution by an economic agent is not free. The participation layer of [section 14](#) separates the spontaneous emergence of a question (dissonance makes the defect salient without any choice) from the deliberate allocation of effort to pursue it (the election ε_i). The former is a cognitive event; the latter is an economic act, and the leverage statistic shows that it can be priced without violating the stricture that pre-insight deliberation must not quantify over unformulated answers.

Causal dissonance should not be identified with inverse insight. A more appropriate later

analogue of inverse insight would be a transition in which the agent grasps that another deterministic customer partition is the wrong kind of explanation—for example, because the relevant law is stochastic, strategic, or located at a different causal level. That transition lies beyond the deterministic four-state model.

17 Results, limitations, and the next theoretical problem

The formal results have different roles. The following are necessary scaffolding:

- (i) private causal beliefs can induce strategically differentiated product commitments;
- (ii) differentiated actions reveal causal labels while pooled actions do not;
- (iii) fixed-space Bayesian conditioning cannot produce a new awareness partition;
- (iv) conflict-free cells characterize restorative refinements; and
- (v) the four-state discovery history has exactly four minimal restorative refinements.

The potentially substantive mechanisms are sharper:

- (i) *Fallibility versus exhaustiveness.* Observationally equivalent worlds preclude a uniformly correct truth-blind formulation, while a representation robustly sufficient for both row and column worlds must be completely fine.
- (ii) *Representation-relative certainty.* A false causal formulation can receive probability one under retrospective renormalization inside its newly produced model space.
- (iii) *Path-dependent awareness.* Firms can converge to the same true causal theory and policy while retaining different awareness structures.
- (iv) *Endogenous diagnosticity.* Theories determine actions, actions determine which causal labels experience reveals, and revealed labels determine whether theories survive.
- (v) *Self-confirming causal homogeneity.* Shared false formulations can align strategies and suppress their own falsification, whereas causal heterogeneity generates both learning and asymmetric value capture.
- (vi) *Delabeled question value.* The subjective value of a question is computable from retained episodes alone, is invariant across every representation that could answer it, and tracks action separation rather than residual model multiplicity. Consequently the awareness trap is robust to cheap inquiry, sustained competitive success rationally extinguishes inquiry, and differentiated defeat makes the losing firm the natural inquirer.
- (vii) *Unwarranted certainty and its invisibility.* Retrospective fit is guaranteed by restoration and hence uninformative; representation- relative certainty exceeds selection-adjusted warrant by a gap that pooled play preserves forever; and no firm short of complete awareness can express the comparison that measures its own gap. Question value and warrant improvement share a currency: both are created by differentiated action and destroyed by pooling.

- (viii) *Knife-edge pathologies*. The self-confirming trap, the permanent warrant gap, and the zero-leverage lock all require demand to absorb misservice completely. With any outside option, consistency and subjective optimality force correct service of every recurring state; false theories persist only as payoff-irrelevant neutral mislabels, and the destroyed value is itself the corrective signal. Diagnosticity has two sources—strategic heterogeneity and demand-side harshness—and collective error requires the absence of both.

The example does not yet supply a complete strategy-field contribution. In particular, it endogenizes only the binary participation margin of inquiry, not its intensity or duration; and it does not endogenize the production of a bridge prior, the accumulation of agent-specific heuristic accessibility, or the organizational and competitive conditions that make alternative inquiry technologies valuable. Section 15 connects false awareness to value creation through the outside option, but the demand response itself remains exogenous; endogenizing β through prices, customer search, or exit dynamics—and characterizing equilibrium in the anti-coordination region where no constant action is dominant—remains open.

The next theoretical problem is therefore not simply to enlarge X . It is to endogenize the representation-producing technology and its strategic environment while preserving the four-state model’s central discipline: experience may direct inquiry, but neither inquiry nor Bayes may be allowed to select the objective truth by assumption.

18 Conclusion

The 2×2 environment isolates a formal boundary between learning within a theory space and producing a new theory space. Strategic differentiation creates a causal conflict that no initially expressible theory can resolve. Restoration and parsimony narrow the possible responses from fifteen partitions to two simple factor-based candidates, but observational equivalence prevents the experience from selecting the true one. A non-exhaustive inquiry technology must therefore be fallible. Once a formulation is produced, fixed-space Bayesian assessment can resume, but representation-relative certainty must be separated from judgment based on adequate fresh or selection-adjusted evidence. Finally, strategic action feeds back into the epistemic process: shared false formulations may become self-confirming by eliminating diagnostic variation, whereas theory heterogeneity generates both competitive rents and the experience required for correction. Both sides of that contrast presuppose perfectly forgiving demand; with any outside option, the customer side supplies the discipline that pooled strategies suppress.

References

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